

Wouter van Amsterdam, MD, PhD

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Positions

from 2023	University Medical Center Utrecht , assistant professor
2021 – 2023	Babylon Health, London , senior research scientist

Degrees

2022	Utrecht University , PhD, <i>Causal Inference and Machine Learning for Health</i> , co-advised by Rajesh Ranganath, New York University, partially conducted at NYU in 2019
2020	University Medical Center Utrecht , MSc. Epidemiology, medical statistics track, <i>cum laude</i>
2017	University Medical Center Utrecht , MD (honours track, master thesis at Harvard Medical School)
2014	University Medical Center Utrecht , BSc. in Medicine (honours track)
2010	Utrecht University , BSc. in Physics and Astronomy

Selected Publications

Clinical trials for continuously monitored and updated AI systems.

Van Amsterdam, W. A. C., Michael Oberst, Jean Feng, et al. Nat Med. 2026 Apr 28. doi:10.1038/s41591-026-04368-9.

The Enemies of Reliable and Useful Clinical Prediction Models: A Review of Statistical and Scientific Challenges. Annual Review of Statistics and Its Application. 2025 Aug 28.

Van Calster B, Van Smeden M, **van Amsterdam W.A.C.**, Coemans M, Wynants L, Steyerberg EW.

The Risks of Risk Assessment: Causal Blind Spots When Using Prediction Models for Treatment Decisions.

Van Geloven N, Keogh RH, **van Amsterdam W.A.C.**, Cinà G, Krijthe JH, Peek N, et al. Ann Intern Med. 2025 Sept;178(9):1326–33.

AI as an intervention: improving clinical outcomes relies on a causal approach to AI development and validation

Shalmali Joshi, Inigo Urteaga, **van Amsterdam, W.A.C.** et al. JAMIA 2025.

When accurate prediction models yield harmful self-fulfilling prophecies.

van Amsterdam, W. A. C., van Geloven, N., Krijthe, J. H., Ranganath, R., & Cinà, G. Patterns. 2025 Apr;6(4):101229.

Prognostic models for decision support need to report their targeted treatments and the expected changes in treatment decisions.

van Amsterdam, W. A. C. and Cinà, Giovanni and Didelez, Vanessa and Keogh, Ruth H. and Peek, Niels and Sperrin, Matthew and Vickers, Andrew J. and van Geloven, Nan and Shalit, Uri (2024). BMJ Rapid-response

From algorithms to action: improving patient care requires causality.

van Amsterdam, W. A. C., de Jong, P. A., Verhoeff, J. J. C., Leiner, T., & Ranganath, R. (2024). BMC Medical Informatics and Decision Making

Causal Inference in Oncology: Why, What, How and When.

van Amsterdam W. A. C. , Elias S, Ranganath R. Clinical Oncology. 2024 Jul 11;

Individual treatment effect estimation in the presence of unobserved confounding using proxies: A cohort study in stage III non-small cell lung cancer.

van Amsterdam, W. A. C., Verhoeff, J. J. C., Harlianto, N. I., Bartholomeus, G. A., Puli, A. M., de Jong, P. A., Leiner, T., van Lindert, A. S. R., Eijkemans, M. J. C., & Ranganath, R. (2022). Scientific Reports

Conditional average treatment effect estimation with marginally constrained models.

van Amsterdam, W. A. C., & Ranganath, R. (2023). Journal of Causal Inference

Eliminating biasing signals in lung cancer images for prognosis predictions with deep learning.

van Amsterdam, W. A. C., Verhoeff, J. J. C., de Jong, P. A., Leiner, T., & Eijkemans, M. J. C. (2019). Npj Digital Medicine

The morphological substrate for Renal Denervation: Nerve distribution patterns and parasympathetic nerves. A post-mortem histological study.

van Amsterdam W. A. C., Blankestijn P. J., Goldschmeding R., Bleys R. L. A. W.. Annals of Anatomy - Anatomischer Anzeiger 2016 Mar;204:71–9

Pre-prints

Causally-interpretable meta-analysis using aggregate data. arXiv preprint

Shi Q, **van Amsterdam, W. A. C.**, Gemert S la B van, Feenstra T, Dahabreh IJ. (2026). arXiv:2605.27272

ECG classification with convolutional neural networks demonstrates resilience to sex-imbalances in data.

Schipaanboord DJ, Van Der Zalm F, Van Es R, Vessies M, Van De Leur RR, Siegersma KR, ... **van Amsterdam W.A.C.**. (2025) MedArXiv preprint

A causal viewpoint on prediction model performance under changes in case-mix: discrimination and calibration respond differently for prognosis and diagnosis predictions.

van Amsterdam, W. A. C. (2024). arXiv:2409.01444

Teaching

As Advisor (2022-onwards)

- 2025-: PhD Student (SciML4Medicine), copromotor, PI
- 2025-: PhD Student (Causal Inference in Radiotherapy), copromotor (remote supervision, INT Milan)
- 2025-: PhD Student (Prediction Paradox in Clinical Prediction Models), copromotor
- 2025-: MBA Student (AI), thesis advisor
- 5+ MSc. students (Data Science, Epidemiology, Medicine)

As lecturer

2026-	Foundations of Statistics and Data Science (Epidemiology and Health Data Science MSc.)
2026-	Machine Learning & Application in Medicine (Epidemiology and Health Data Science MSc.)
2024-	Introduction to Causal Inference and Causal Data Science summer school (yearly)
2023, 2024	2024 Big Data summer school

Awards & grants

2026	ZonMW Veni (€320,000) - Target Trial Assistant: a hybrid deterministic / LLM system to improve methodological quality and accessibility of causal inference in medicine
2026	Accelerator Grant (€120,000) - two year project on Scalable Causal Inference - Mining DAGs from published research & applying in large cohorts
2025	Applied Data Science, Utrecht University, Boost Grant (€5,000) for DagMiner: extracting DAGs from published research
2023	HealthHolland TKI Public-Private partnership (principal investigator), €300,000
2020	Frits de Waard Penning, Epidemiology Master Utrecht, For the best and most original epidemiological research by a student in medicine or biomedical/health sciences, €500
2019	Stichting Drie Lichten - Travel grant for visiting NYU during my PhD, €5000
2019	Stichting Girard De Miele van Coehoorn - Travel grant for visiting NYU during my PhD, €1250
2018	ZonMW, Good use of medicine grant (Goed Gebruik Geneesmiddelen), co-applicant, €650,000 (95,000 for my PhD)
2018	NVIDIA - GPU academic seeding program, Graphics processing unit for deep learning experiments (value = €12,000)
2017	Alexandre Suerman Fellowship MD / PhD program, PhD stipend and masterclass program (value = €180,000)
2016	Bright Minds award of Utrecht University, Pitch presenter & winner at the <i>Dare to Cross-Over</i> event on <i>Diagnosing lung cancer with Radiomics</i> , €500
2015	BBMRI-NL research grant, co-applicant for project 'Systems Radiology', €20,000

Professional activities and memberships

Selected presentations and invited seminars

2026: keynote speaker for *Causal Machine Learning in Medicine, Munchen (invite only workshop)*; invited (visiting) speaker at *INRIA Montpellier*; **2025:** *ISCB* - invited speaker in session 'Prediction modelling meets causal inference for clinical decision making'; *Dagstuhl seminar* - invited seminar participant and speaker; **2024:** *ESTRO (radiotherapy)* - **invited** seminar on causal inference; *AIBIA Utrecht* - **invited** talk; **2023:** *University of Manchester* - **invited** on-site seminar; *CHIL conference* - lightning talk; *MLHC* - **invited** seminar in pre-conference workshop; *ML4H* - poster-presentation oral talk (missed due to canceled flight) **2022:** *Lorentz workshop Leiden* **invited** seminar **2021:** *MLHC* - poster presentation; *ESTRO* - poster presentation; *Dutch association of epidemiology yearly symposium* - **invited** seminar (canceled due to COVID lockdown) **2019:** *AAAI-Why symposium* - oral abstract presentation **2018:** *Probus Service Club Utrecht - **invited** seminar

Affiliations and memberships

- general program chair, [MLHC 2026 & 2027](#)
- program chair Workshop, [MLHC 2025](#)
- board member [BMS-ANed](#) (Dutch biometrics society, board member)
- coordinator of UMC Utrecht [AI methods lab](#)
- ambassador for [Applied Data Science](#) of Utrecht University
- co-coordinator of [Causal Data Science Special Interest Group](#) of Utrecht University

Reviewing & Editing

2026 associate editor, Diagnostic and Prognostic Research *2025-2026* guest editor, special issue *ISCB46 2024-2025* guest editor, special issue on AI, Diagnostic and Prognostic Research *2024* statistical editor, Journal of Cardiovascular Magnetic Imaging

Biostatistics; Machine Learning for Healthcare @NeuRIPS (2019, 2020, 2021, 2022); IEEE-access; Journal of Cardiovascular Magnetic Resonance; Netherlands Heart Journal; Oncologie; Journal of Thoracic Disease; Spurious correlations, Invariance and Stability workshop @ ICML 2022 (program committee); Translational Lung Cancer Research, Biometrika

Student Research Positions

2019 visiting PhD student at CILVR, New York University, with Rajesh Ranganath (4 months); **2016** master thesis at Harvard Medical School, with Hugo Aerts (4 months); **2016** master honours program at Radiology department of University Medical Center Utrecht (1.5 years); **2014** bachelor honours program at Anatomy department of University Medical Center Utrecht (2.5 years); **2010** bachelor thesis at *Physics of Man*, Utrecht University (3 months);

In the News

- Our work on harmful-selffulfilling prophecies got covered in [The Independent](#), [Pharmophorum](#), and 6 independent experts at an [SMC-roundup](#)
- I gave an interview for BiotechNEWS on my vision for the future of AI in healthcare [pdf](#)

Languages

Dutch, Native; English & Speak fluently and write with high proficiency; German & French & Basic proficiency
R programming language & Fluently; Python programming language & Fluently, experience in PyTorch, Jax, Pyro, NumPyro (contributor)